Multiple Sequence Alignment (MSA)

- Sequence alignment is a way of arranging the sequences of DNA, RNA, or protein to identify regions of similarity
- Ex: Human/zebrafish (What does the local alignment of two dots mean?)
- Across the living organisms this rule is true. Proteins performing the same function in different organisms exhibit similar amino acid or gene sequences.
- These similarities generally refer to functional equivalence and evolutionary relationships between two proteins.
- A multiple sequence alignment is basically an alignment of more than 2 sequences
- A pairwise alignment - tells you about the similarity of the sequences
- A multiple alignment - tells you about the similarity among multiple sequences
- Pairwise -> Water [EMBOSS] (local), Needle [EMBOSS] (global)
- MSA -> Muscle, Clustalw, MAFFT, T-Coffee (local), Block Maker (global)
- Local Alignment aims to identify small highly similar regions of two proteins or genes (minimum gaps)
- Global prioritizes end-to-end alignment of the sequences, reflecting overall variations
- Dynamic Programming Approach
  ↳ This is an extension of dynamic programming approach for pairwise alignment to multiple sequences.
  ↳ Unfortunately the cost of computation grows exponentially (Not very effective)
- Progressive Alignment Method
  ↳ Multiple variants -> i.) differ in how to choose the order to do alignment, ii.) how the progression involves inclusion of sequences to a growing alignment, and iii.) the procedure used to align and score the alignments.
- Iterative Refinement Method
  ↳ Similar to progressive alignment but when a new sequence is added all sequences are repeatedly realigned in order to find the most optimal alignment.
- Uniprot + Clustal Omega for exact hands-on protein analysis (+ command prompt)
- Protein sequences are formatted in FASTA sequence files
- Download -> Clustal -> Use command prompt -> Open FASTA in Ugene -> Alignment + [illegible]
  ↳ How? -> Step 1 -> Alignment between all sequence pairs (pairwise alignment) (Pairwise Alignment)
  ↳ Step 2 -> Distance matrix -> Step 3 -> Creates a guide tree (more similar two sequences -> 2 to [illegible])
  ↳ Clustering algorithm, creates a cladogram
- Distance values that are smaller signify greater levels of similarity
- Distance matrix ≠ Covariance matrix (Though both are symmetric) Also all covariance matrices are PSD matrices.